

Justin Jiang

[linkedin.com/in/jusjiang](https://www.linkedin.com/in/jusjiang) | github.com/nitsujiang | (347) 865-7284 | jiangjustin.jj@gmail.com

EDUCATION

Cornell University

M.Eng. in Computer Science

Ithaca, NY

Expected May 2027

Stony Brook University

B.S. in Computer Science | GPA: 3.81/4.00

Stony Brook, NY

May 2026

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript/TypeScript, SQL, HTML/CSS

Frameworks & Libraries: React, Node.js, Express, Next.js, Django, Tailwind CSS, LangChain, Pandas, NumPy, scikit-learn, Matplotlib

Databases: MongoDB, PostgreSQL

Tools: Docker, Redis, Prometheus, Grafana

PROFESSIONAL EXPERIENCE

Research Assistant – COMPAS Lab

Stony Brook University

January 2026 – May 2026

Stony Brook, NY

- Built a Python CLI tool automating local setup of a distributed genomic search system — provisioning isolated coordinator/worker environments and containerized MongoDB instances with a single command — replacing a fragile multi-step manual workflow
- Diagnosed a data corruption bug in the system's probabilistic indexing structure caused by integer underflow during high-load merge operations, and patched the underlying edge case
- Refactored client-service RPC handling into a single centralized communication layer, replacing scattered per-component call logic with one consistent interface for coordinating distributed components

Machine Learning Fellow

Break Through Tech AI

May 2025 – April 2026

New York, NY

- Completed a 12-month fellowship combining ML coursework, an industry-level project, mentorship, and an agentic AI specialization
- Built a Streamlit dashboard under Snowflake engineers, training classification models on weather, hospital, and social vulnerability data to flag medical-desert risk across Arizona, surfacing heat-risk and hospital capacity as leading drivers

Systems Programming Teaching Assistant

Stony Brook University

August 2024 – December 2024

Stony Brook, NY

- Led weekly recitations and office hours on C programming and systems debugging for 100+ students; collaborated with course staff to design problem sets and exam questions

PROJECTS

Discord Clone | *TypeScript, PostgreSQL, WebSockets, Docker, Redis* | [GitHub](#)

March 2026 – May 2026

- Deployed a real-time messaging platform on an 8-node Docker Swarm cluster, supporting 4K peak concurrent connections at 240+ req/s
- Identified and resolved lock contention between GIN index updates for full-text search and message insert paths by decoupling search indexing onto async background workers, dropping insert latency from 200ms to <1ms
- Built a distributed Redis caching layer for high-frequency read-state updates, preventing PostgreSQL connection saturation under load

TaskPilot | *Python, Gemini API, Slack SDK, Google Workspace* | [Github](#)

January 2026 – April 2026

- Built a Slack-integrated AI agent using the Gemini API to extract action items, owners, and deadlines, triggering clarification prompts when the model's confidence or field-extraction completeness indicated ambiguity rather than silently guessing
- Added meeting-scheduling workflows with Google Calendar conflict checks and Gmail reminder delivery

C-Like Compiler | *C++, LLVM* | [Github](#)

September 2025 – December 2025

- Built a compiler for a C-like language with a recursive descent parser, AST generation, and control flow via right-recursive grammar transformations
- Developed a semantic analyzer with a scoped symbol table for type checking and identifier resolution, and integrated LLVM IR code generation
- Implemented on-the-fly SSA construction with recursive predecessor search, incomplete-phi tracking, and block-sealing to correctly resolve variable definitions across loops and conditional branches